NETWORK AND CONTENT DELIVERY

Networking enables devices, applications, and services to communicate and exchange data. In the AWS Cloud, networking services provide the foundation for connecting cloud resources, controlling how traffic moves between them, and making applications accessible to users. Understanding basic networking concepts is therefore important before configuring network resources in AWS.

This module introduces networking fundamentals and the AWS services used to build and deliver cloud-based networks. It covers Amazon Virtual Private Cloud (Amazon VPC), VPC networking and security, Amazon Route 53, and Amazon CloudFront, providing the foundation for creating a virtual private cloud and controlling connectivity between AWS resources and external networks.

Networking Basics

A **computer network** consists of two or more devices connected so that they can communicate and share information. Devices connected to a network use established rules and addressing mechanisms to determine where information should be sent and how it reaches its destination.

Networks can vary in size and purpose. A small network might connect devices within a limited location, while larger networks can connect systems across different geographic areas. Regardless of size, network communication depends on identifying devices and directing traffic between a source and a destination.

Internet Protocol Addresses

An Internet Protocol (IP) address is a numerical address used to identify a device on a network. IP addresses allow network traffic to be directed to the appropriate destination.

Internet Protocol version 4 (IPv4) addresses contain four decimal numbers separated by periods. Each number represents 8 bits, producing a **32-bit address**. An example of an IPv4 address is 192.168.1.10.

IP addresses can be public or private. A **public IP address** can be used for communication over the internet, while **private IP addresses** are used for communication within private networks. Private IPv4 address ranges include:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

IPv4 and IPv6

The number of possible IPv4 addresses is limited because IPv4 uses a 32-bit address space. **Internet Protocol version 6 (IPv6)** was developed to provide a much larger address space and uses **128-bit addresses**.

IPv6 addresses are written as groups of hexadecimal values separated by colons. The larger address space provides substantially more unique addresses than IPv4 and supports the continued growth of internet-connected devices and services.

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Classless Inter-Domain Routing

Classless Inter-Domain Routing (CIDR) is a method used to represent a network and specify the range of IP addresses available within it. CIDR notation combines an IP address with a prefix length, such as 10.0.0.0/16.

The number following the slash indicates how many bits identify the **network portion** of the address. The remaining bits are available for identifying addresses within that network. For example, a /16 prefix uses the first 16 bits for the network portion, while a /24 prefix uses the first 24 bits. As the prefix length increases, the range contains fewer IP addresses.

Understanding CIDR is particularly important when creating an Amazon VPC because an IP address range is assigned to the VPC and portions of that range can later be assigned to its subnets.

The Open Systems Interconnection Model

The Open Systems Interconnection (OSI) model organizes network communication into seven layers. Each layer represents a particular set of networking functions and works with other layers to support communication between systems.

The seven layers are:

1. **Physical** – Transmits raw bits through physical media.
2. **Data Link** – Supports communication between devices on the same network link.
3. **Network** – Provides logical addressing and routing between networks.
4. **Transport** – Manages end-to-end communication and data delivery.
5. **Session** – Establishes and manages communication sessions.
6. **Presentation** – Handles the representation and formatting of data.
7. **Application** – Provides network services used by applications.

The OSI model provides a structured way to understand how network communication occurs. In cloud networking, concepts such as IP addressing, routing, ports, protocols, and network security controls operate at different layers and become important when configuring an Amazon VPC and its resources.

Amazon Virtual Private Cloud (Amazon VPC)

Amazon Virtual Private Cloud (Amazon VPC) is an AWS service that enables customers to provision a logically isolated virtual network in the AWS Cloud. Within a VPC, AWS resources can be launched into a network environment where customers define important networking settings, including IP address ranges, subnets, routing, and connectivity.

A VPC belongs to a single AWS Region but can span multiple Availability Zones (AZs) within that Region. This allows resources to be distributed across Availability Zones while remaining part of the same virtual network.

VPC IP Addressing

When creating a VPC, an IP address range is assigned to it using **Classless Inter-Domain Routing (CIDR)** notation. This CIDR block establishes the range of IP addresses that can be used within the VPC.

The VPC's address range should be planned according to the number of resources and subnets that the network will need. Portions of the VPC CIDR block can then be allocated to individual subnets. Because the address ranges of the subnets come from the VPC address range, their CIDR blocks must not overlap.

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Subnets

A **subnet** is a range of IP addresses within a VPC. While a VPC spans the Availability Zones of a Region, each subnet resides entirely within one Availability Zone.

Subnets allow resources within a VPC to be organized into smaller network segments. For example, an architecture can place resources that require direct internet connectivity in one subnet while keeping other resources in a subnet without direct internet access.

Subnets are commonly described as:

- **Public subnet** – A subnet whose route table has a route to an internet gateway, allowing resources with appropriate addressing and configuration to communicate with the internet.
- **Private subnet** – A subnet whose route table does not have a route to an internet gateway. Resources in the subnet are not directly accessible from the internet through an internet gateway.

A subnet itself does not automatically make every resource public or private. Routing, IP addressing, and security configuration also determine whether a particular resource can communicate with the internet.

AWS reserves five IP addresses in each IPv4 subnet CIDR block. The first four IP addresses and the last IP address are not available for assignment to resources. For example, a /24 subnet contains 256 IP addresses, of which 251 are available for use after the five reserved addresses are excluded.

Public and Private IP Addresses

AWS resources within a VPC can use **private IP addresses** for communication within the VPC and connected private networks. A private IPv4 address is assigned from the address range of the subnet in which the resource is located.

Some resources can also be assigned a **public IPv4 address** when internet communication is required. Public IPv4 addresses enable communication through the internet when the necessary routing and security settings are also configured.

AWS also provides **Elastic IP addresses**, which are static public IPv4 addresses that can be associated with supported AWS resources. Unlike an automatically assigned public IPv4 address, an Elastic IP address can remain allocated to an AWS account until it is released.

VPC Components

An Amazon VPC can include several networking components that work together to control connectivity and traffic flow. These include subnets, route tables, internet gateways, network access control lists (network ACLs), and security groups.

Each component performs a different role. Subnets divide the VPC into network segments, route tables determine where network traffic is directed, and gateways provide connectivity between the VPC and other networks. An **elastic network interface** is a logical networking component in a VPC that represents a virtual network card and can include attributes such as private IP addresses. Security groups and network ACLs provide controls for traffic entering or leaving resources and subnets.

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These components establish the basic structure of a VPC and provide the foundation for controlling communication within the VPC and with external networks.

VPC Networking

After a VPC and its subnets are created, network traffic needs a way to move between resources and reach destinations outside the VPC when required. **VPC networking** uses components such as route tables, internet gateways, and network address translation to determine where traffic can travel.

The way these components are configured affects whether resources can communicate within the VPC, connect to the internet, or remain isolated from direct internet access.

Route Tables
A **route table** contains a set of rules, called **routes**, that determine where network traffic from a subnet or gateway is directed. Each route specifies a destination and a target for traffic that matches that destination.

Every subnet in a VPC must be associated with a route table. If a subnet is not explicitly associated with a custom route table, it uses the VPC's main route table.

A route table includes a local route that enables communication within the VPC. Additional routes can be added when traffic needs to reach destinations outside the VPC.

Internet Gateways
An **internet gateway** is a VPC component that enables communication between a VPC and the internet. It is attached to a VPC and can serve as a target in a route table for internet-bound traffic.

For a resource in a public subnet to communicate with the internet through an internet gateway, the necessary network configuration must be present. This generally includes:

- An internet gateway attached to the VPC
- A route in the subnet's route table that directs internet-bound traffic to the internet gateway
- An appropriate public IP address for the resource
- Security rules that permit the required traffic

A common IPv4 route for internet-bound traffic uses **0.0.0.0/0** as the destination and the internet gateway as the target. This route represents IPv4 destinations outside the more specific routes in the route table.

Network Address Translation
Resources in a private subnet might need to initiate connections to services outside the VPC without accepting unsolicited inbound connections directly from the internet. **Network Address Translation (NAT)** can be used to support this type of outbound connectivity.

A **NAT gateway** can allow resources in a private subnet to connect to destinations outside the VPC while preventing those external systems from initiating connections through the NAT gateway to the private resources.

In a typical configuration, the private subnet's route table directs internet-bound traffic to a NAT gateway. The NAT gateway is placed in a public subnet, which has a route to an internet gateway.

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Public and Private Subnet Routing

Routing determines how resources in public and private subnets communicate with destinations outside the VPC. Resources in a public subnet can use a route to an internet gateway when direct internet connectivity is required. Resources in a private subnet can instead use a NAT gateway when they need to initiate outbound connections without allowing unsolicited inbound connections directly from the internet.

This configuration allows an application to separate resources according to their connectivity requirements. For example, a public-facing web server can be placed in a public subnet, while resources that do not require direct internet access can remain in private subnets.

VPC Security

A VPC can contain resources with different access requirements. Some resources might need to receive traffic from the internet, while others should accept traffic only from specific systems. Amazon VPC provides security controls that help regulate inbound and outbound network traffic. Two fundamental controls are security groups and network access control lists (network ACLs).

Although both can filter network traffic, they operate at different levels and behave differently. Security groups control traffic associated with resources such as Amazon EC2 instances, while network ACLs control traffic at the subnet level.

Security Groups

A security group acts as a virtual firewall that controls inbound and outbound traffic for associated resources. In the module, security groups are described as operating at the instance level rather than the subnet level, which allows instances within the same subnet to use different security groups.

Security group rules specify the traffic that is permitted. Rules can consider information such as the protocol, port range, and source or destination. For example, a web server might permit inbound HTTP traffic on port 80 and HTTPS traffic on port 443 while restricting administrative access such as SSH to an appropriate source.

Security groups support allow rules, but not explicit deny rules. When a security group is created, it initially has no inbound rules, so inbound traffic originating from another host is not permitted until appropriate rules are added. By default, a security group includes a rule that allows outbound traffic.

Stateful Traffic Filtering

Security groups are **stateful**. This means that when traffic is permitted in one direction, the corresponding response traffic is automatically permitted regardless of the rules for the opposite direction.
For example, if an instance sends an allowed request to another system, the response to that request can return to the instance without requiring a separate inbound rule specifically for the response traffic. Similarly, responses to allowed inbound requests can leave the instance regardless of outbound rules.

Network Access Control Lists

A **network access control list (network ACL)** is an optional security layer that controls traffic entering and leaving one or more subnets. Unlike security groups, which operate at the instance level, network ACLs operate at the subnet level.

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Each subnet must be associated with a network ACL. If a subnet is not explicitly associated with another network ACL, it is associated with the default network ACL. A network ACL can be associated with multiple subnets, but a subnet can be associated with only one network ACL at a time.

A network ACL contains separate inbound and outbound rules. Unlike security groups, its rules can explicitly allow or deny traffic. Rules are numbered and evaluated in order, beginning with the lowest numbered rule that matches the traffic.

Stateless Traffic Filtering
Network ACLs are **stateless**, meaning that information about a request is not retained after the request is processed. Therefore, allowing traffic in one direction does not automatically permit the corresponding response traffic in the opposite direction. Appropriate rules must be configured for both directions when necessary.

The **default network ACL** allows inbound and outbound traffic. In contrast, a newly created **custom network ACL** initially denies inbound and outbound traffic until rules are added to permit the required communication.

Security Groups and Network ACLs
The differences between the two controls are important when designing VPC security:

Attribute	Security Groups	Network ACLs
Scope	Instance/resource level	Subnet level
Rules	Allow rules	Allow and deny rules
State	Stateful	Stateless
Rule processing	All applicable rules are evaluated	Rules are evaluated in number order

Security groups and network ACLs can therefore provide different layers of traffic control within a VPC. Security groups protect associated resources, while network ACLs provide an additional control at the subnet boundary.

Amazon Route 53

Amazon Route 53 is a highly available and scalable **Domain Name System (DNS)** web service. DNS translates human-readable domain names, such as www.example.com, into IP addresses that computers use to communicate with one another. Route 53 can connect user requests to infrastructure running in AWS and can also route users to infrastructure outside AWS.

In addition to DNS routing, Amazon Route 53 supports domain registration and health checking. These capabilities allow organizations to register domain names, direct users to application resources, and monitor the health of resources used to respond to requests.

Domain Name System
The Domain Name System (DNS) provides a way to associate domain names with IP addresses. Instead of requiring users to remember an IP address, DNS enables them to access a resource through a recognizable domain name.

When a user enters a domain name, a DNS resolver begins the process of locating the corresponding address. The request can pass through DNS servers until the authoritative DNS server provides the required

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information. The resulting IP address is returned to the user's device, which can then connect to the requested resource.

Route 53 Routing Policies

Amazon Route 53 provides several routing policies that determine how it responds to DNS queries. The appropriate policy depends on how traffic should be directed among available resources.

- **Simple routing** – Routes traffic to a single resource that performs a given function.
- **Weighted routing** – Routes specified proportions of traffic to multiple resources.
- **Latency routing** – Routes traffic based on the AWS Region that provides the lowest latency for the user.
- **Failover routing** – Directs traffic to an alternative resource when a primary resource becomes unavailable.
- **Geolocation routing** – Routes traffic according to the geographic location from which DNS queries originate.
- **Geoproximity routing** – Routes traffic according to the geographic location of resources and users.
- **Multivalue answer routing** – Returns multiple values, such as IP addresses, in response to DNS queries.

Health Checks and Failover

Amazon Route 53 can perform **health checks** to determine whether resources are available and functioning. Health information can be combined with DNS routing to direct traffic away from unhealthy resources.

For example, an application can operate in more than one AWS Region. Route 53 can direct users to the primary deployment while monitoring its health. If the primary resource becomes unavailable, a **failover routing policy** can direct DNS queries to a secondary resource. This approach can help applications remain accessible when a resource or deployment experiences a failure.

Together, DNS routing, health checking, and multiple routing policies make Amazon Route 53 an important service for directing users to available application resources.

Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) service that delivers data, applications, and other content to users with low latency and high transfer speeds. It uses a worldwide network of edge locations to place content closer to users, reducing the distance that requests and responses need to travel.

CloudFront can deliver both static content, such as images and files, and dynamic content, such as application-generated responses. Frequently requested content can be cached at locations closer to users so that it does not always need to be retrieved from the original source.

Content Delivery Networks

A **content delivery network (CDN)** is a geographically distributed system that helps deliver content closer to users. Without a CDN, a user's request might need to travel a long distance to reach the server where the content is hosted. This distance can increase **latency**, or the delay experienced when transmitting data across a network.

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A CDN reduces this delay by storing copies of content at locations closer to users. When requested content is available in a nearby cache, it can be returned without retrieving it from the original server.

Edge Locations and Regional Edge Caches

Amazon CloudFront uses edge locations and Regional edge caches as part of its global infrastructure. Edge locations are positioned closer to users and can cache content that is frequently requested.

When a user requests content through CloudFront, the request is directed to an appropriate edge location. If the requested content is already stored in the cache, CloudFront can deliver it directly to the user. If it is not available, CloudFront retrieves the content from the configured origin, delivers it to the user, and can cache it for later requests.

CloudFront Origins

An **origin** is the location where the original version of content is stored. CloudFront retrieves content from an origin when the requested object is not available in the appropriate cache.

AWS resources can serve as origins for a CloudFront distribution. This enables CloudFront to work with application and storage resources while using its distributed network to deliver their content to users.

Benefits of Amazon CloudFront

Amazon CloudFront provides several benefits for content delivery:

- **Fast** – Delivers content through a globally distributed network that places content closer to users.
- **Global** – Uses AWS edge infrastructure to serve users from locations around the world.
- **Secure** – Provides capabilities for securely delivering content and integrates with AWS security services.
- **Scalable** – Uses AWS infrastructure to accommodate changing levels of content requests.
- **Cost-effective** – Uses a pay-as-you-go pricing model rather than requiring customers to build and maintain their own content delivery infrastructure.

By caching content closer to users, Amazon CloudFront can reduce latency and reduce the amount of traffic that must repeatedly reach the origin. Together with Amazon VPC for virtual networking and Amazon Route 53 for DNS and traffic routing, CloudFront supports the delivery of applications and content to users.

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